

PyGameMaker — Tutorial 7: Platformer

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Platformer: Run, Jump, Collect** (4 pages, about 25-30 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A platform game: your character falls with gravity, runs and jumps between platforms, collects coins, avoids spikes (each hit costs a life) and reaches a flag to win.

Phase 1: A Jumping Player

- ☐ 1. Create two sprites (32×32): spr_player and spr_ground.
- ☐ 2. Create obj_ground (sprite spr_ground, **Solid**) and obj_player (sprite spr_player).
- ☐ 3. In obj_player, **Create** event: *Set gravity* direction **270**, strength **0.5**.
- ☐ 4. Add **Keyboard: Left Arrow (held)** with *Set horizontal speed to -4* and **Right Arrow (held)** with *4*.
- ☐ 5. Add **Keyboard: No key** with *Set horizontal speed to 0*. Do **not** use *Stop movement* here: it would also cancel gravity.
- ☐ 6. Add **Key press: Up Arrow** with *Set vertical speed to -10*.
- ☐ 7. Add **Collision with obj_ground** with *Stop movement*.
- ☐ 8. Create room_level1 (800×480, light blue background, Snap to Grid 32×32). Place ground along the bottom, platforms in the air, and the player standing on the ground on the left. **Keep the space above the player free** so it can jump.

You should see: press **F5**. The player falls onto the ground, runs left and right, and the up arrow makes it jump about three tiles high. It lands on platforms.

Phase 2: Coins and Hazards

- ☐ 9. Create sprites spr_coin, spr_spike and spr_flag and objects obj_coin, obj_spike and obj_flag (no events on them).
- ☐ 10. In obj_player add **Collision with obj_coin**: *Add to score 10*, then *Destroy other instance*.
- ☐ 11. Add **Collision with obj_spike**: *Restart room*.
- ☐ 12. Add **Collision with obj_flag**: *Show message* "You Win!".
- ☐ 13. In the room, put coins on platforms, spikes near pits, and the flag at the right end.

You should see: press **F5**. Coins vanish and are worth 10. A spike restarts the level. The flag shows "You Win!".

Phase 3: Game Controller

- ☐ **14.** Create `obj_game_controller` (no sprite).
- ☐ **15. Game Start** event (from **Other Events**): **Set score to 0** and **Set lives to 3**.
- ☐ **16. Draw** event: **Draw score at x 10, y 10** and **Draw lives at x 200, y 10**.
- ☐ **17.** In `obj_player`'s spike collision, add **Add to lives -1** **before** **Restart room**.
- ☐ **18.** Place `obj_game_controller` anywhere in the room.

You should see: press **F5**. Score and lives are shown. Each spike costs one life and restarts the room, and the lives stay lower.

Game Start, not Create. A Create event runs again every time the room restarts. If the lives were set to 3 in Create, every spike would give them back and you would never run out.

Stuck?

What is wrong	Most likely cause	What to do
The player falls through the ground	<code>obj_ground</code> is not Solid , or the player has no Collision with obj_ground	Tick Solid ; add the event with <i>*Stop movement*</i>
The player does not fall at all	The Set gravity action is missing from Create , or No key uses <i>*Stop movement*</i>	Add gravity; change No key to <i>*Set horizontal speed to 0*</i>
The player will not jump	The Up event is <i>*held*</i> instead of <i>*Key press*</i> , or something solid is right above the player	Use Key press: Up Arrow ; keep the space above free
The player jumps again and again while holding Up	It is a Keyboard (held) event	Use Key press so each press is one jump
Coins do nothing	The coin event is on the coin instead of the player, or missing	Put Collision with obj_coin on <code>obj_player</code>
Spikes never cost lives	The lives are set in Create instead of Game Start , or <i>*Add to lives -1*</i> is missing	Move the actions to Game Start ; add the action before <i>*Restart room*</i>
The score and lives do not show	<code>obj_game_controller</code> is not in the room	Place it in the room
The flag does nothing	No Collision with obj_flag on the player	Add it

Challenges

- **Try this (5 minutes):** change the gravity (0.5) or the jump speed (-10) and see how the jump feels.
- **Push further:** add a moving platform or a patrolling enemy; make the flag go to the next room.
- **Invent:** add a double jump, or a Game Over message when the lives reach 0.

Vocabulary

Term	What it means
Gravity	A force that pulls an object down (direction 270) every step
Vertical speed	How fast something moves up (negative) or down (positive)
Key press	An event that happens once when a key goes down
Hazard	Something that hurts the player, like a spike
Game Start	An event that runs once, when the whole game begins

Check Yourself

Write your answers in the notes below.

- 1 Why do we use *Set horizontal speed to 0* and not *Stop movement* for **No key**?
- 2 Why is the jump speed negative?
- 3 Why are the score and lives set in **Game Start** and not in **Create**?

My Notes
